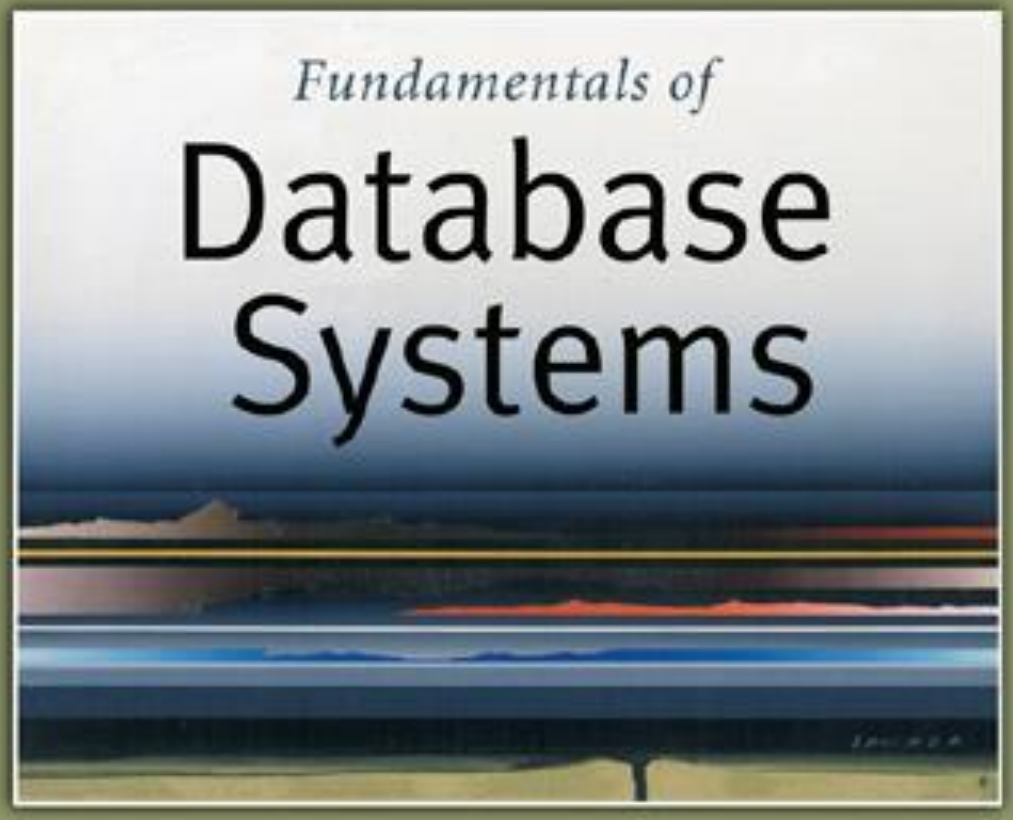

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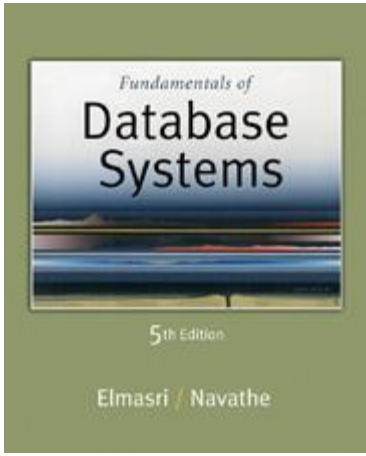
Chapter 14

Indexing Structures for Files



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Chapter Outline

- Types of Single-level Ordered Indexes
  - Primary Indexes
  - Clustering Indexes
  - Secondary Indexes
- Multilevel Indexes
- Dynamic Multilevel Indexes Using B-Trees and B+-Trees
- Indexes on Multiple Keys

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Indexes as Access Paths

- A single-level index is an auxiliary file that makes it more efficient to search for a record in the data file.
- The index is usually specified on one field of the file (although it could be specified on several fields)
- One form of an index is a file of entries <**field value, pointer to record>**, which is ordered by field value
- The index is called an access path on the field.

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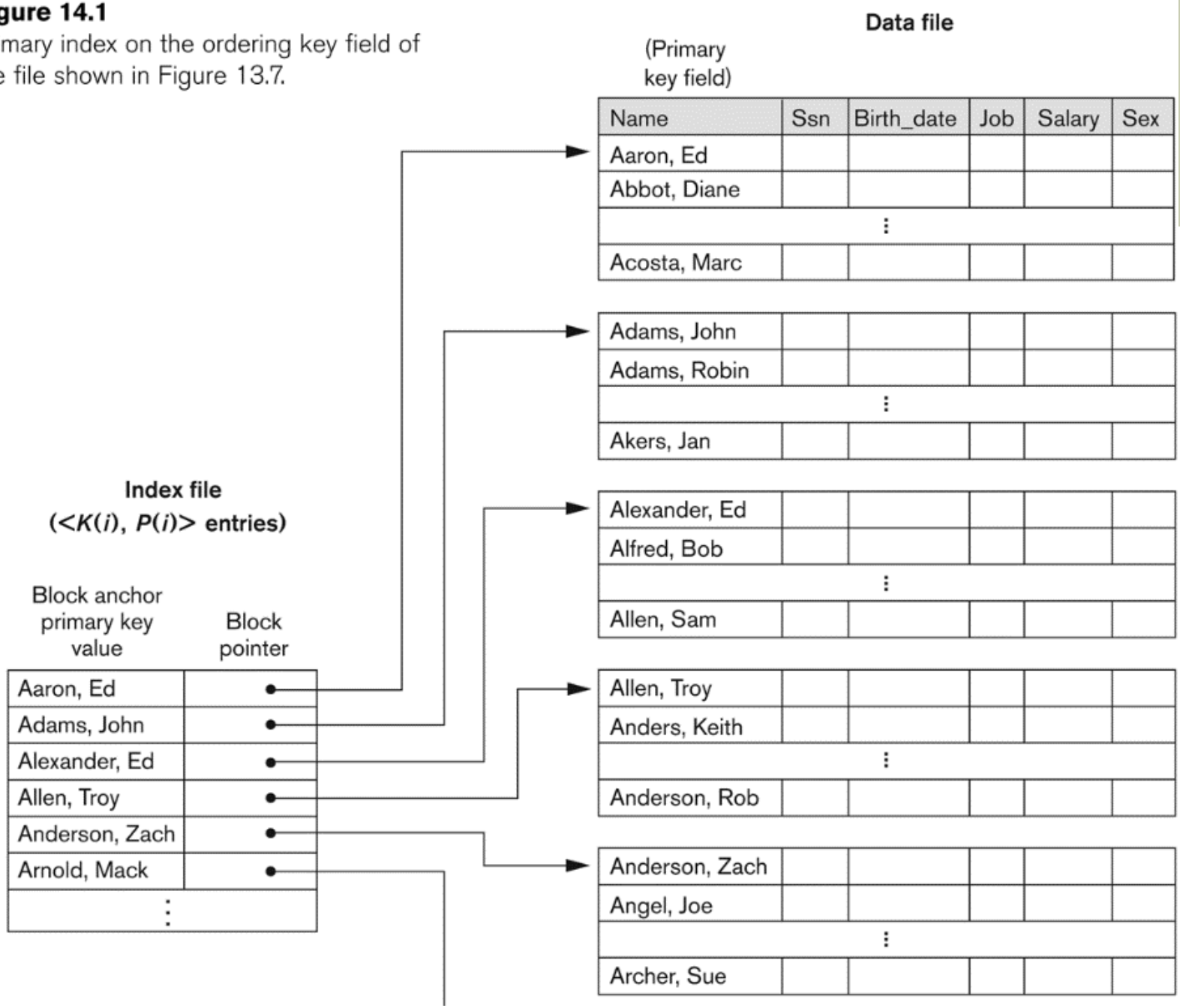
Figure 14.1
Primary index on the ordering key field of the file shown in Figure 13.7.



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Indexes as Access Paths (contd.)

- The index file usually occupies considerably **less disk blocks** than the data file because its entries are much smaller
- A binary search on the index yields a pointer to the file record
- Indexes can also be characterized as dense or sparse
  - A **dense index** has an index entry for every search key value (and hence every record) in the data file.
  - A **sparse (or nondense) index**, on the other hand, has index entries for only some of the search values

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Indexes as Access Paths (contd.)

- Example: Given the following data file EMPLOYEE(NAME, SSN, ADDRESS, JOB, SAL, ...)
- Suppose that:
  - record size R=150 bytes block size B=512 bytes r=30000 records
- Then, we get:
  - blocking factor Bfr= B div R= 512 div 150= 3 records/block
  - number of file blocks b= (r/Bfr)= (30000/3)= 10000 blocks
- For an index on the SSN field, assume the field size V_{SSN}=9 bytes, assume the record pointer size P_R=7 bytes. Then:
  - index entry size R_I=(V_{SSN}+ P_R)=(9+7)=16 bytes
  - index blocking factor Bfr_I= B div R_I= 512 div 16= 32 entries/block
  - number of index blocks b= (r/ Bfr_I)= (30000/32)= 938 blocks
  - binary search needs $\log_2 b = \log_2 938$= 10 block accesses
  - This is compared to an average linear search cost of:
    - (b/2)= 30000/2= 15000 block accesses
  - If the file records are ordered, the binary search cost would be:
    - $\log_2 b = \log_2 30000$= 15 block accesses

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Types of Single-Level Indexes

- Primary Index
  - Defined on an ordered data file
  - The data file is ordered on a **key field**
  - Includes one index entry *for* ***each block*** in the data file; the index entry has the key field value for the *first record* in the block, which is called the ***block anchor***
  - A similar scheme can use the *last record* in a block.
  - A primary index is a nondense (sparse) index, since it includes an entry for each disk block of the data file and the keys of its anchor record rather than for every search value.

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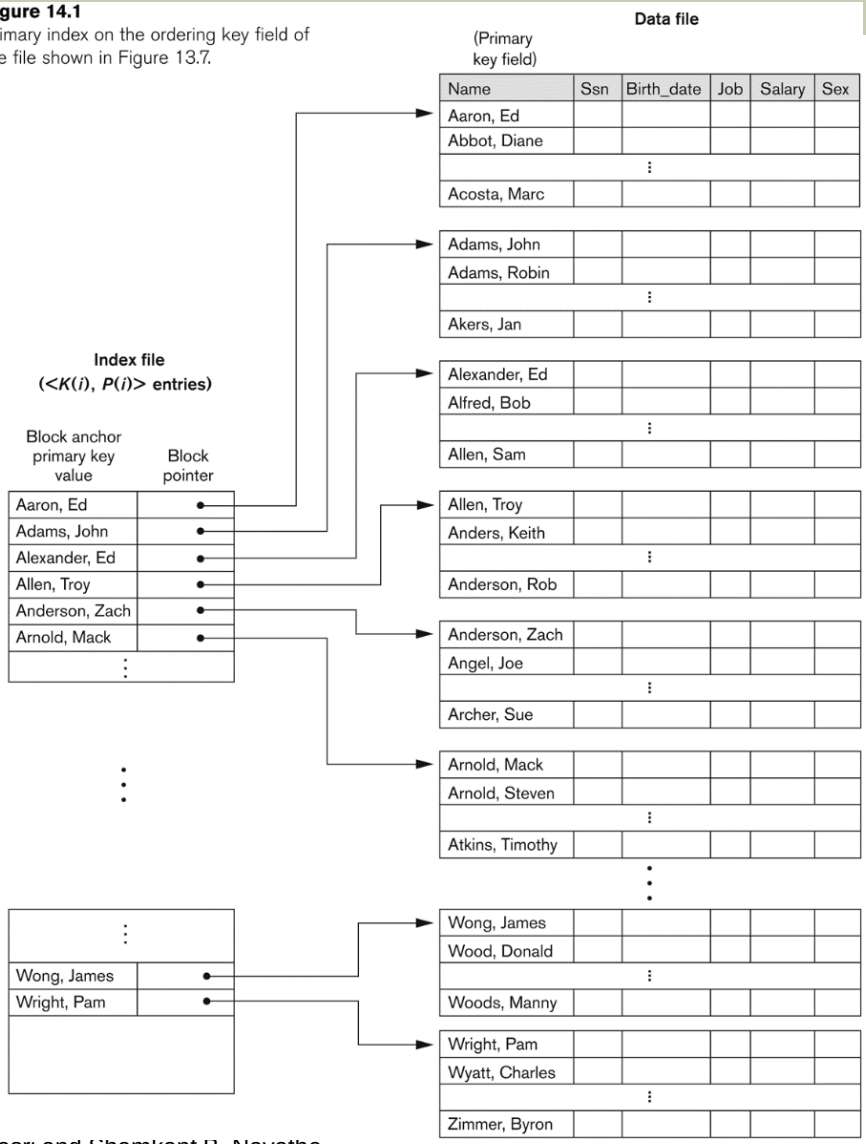
Primary index on the ordering key field

Figure 14.1
Primary index on the ordering key field of the file shown in Figure 13.7.



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Types of Single-Level Indexes

- Clustering Index
  - Defined on an **ordered** data file
  - The data file is ordered on a ***non-key field*** unlike primary index, which requires that the ordering field of the data file have a distinct value for each record.
  - Includes one index entry *for* ***each distinct value*** of the field; the index entry points to the first data block that contains records with that field value.
  - It is another example of *nondense* index where Insertion and Deletion is relatively straightforward with a clustering index.

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A Clustering Index Example

- FIGURE 14.2
  A clustering index on the DEPTNUMBER ordering non-key field of an EMPLOYEE file.

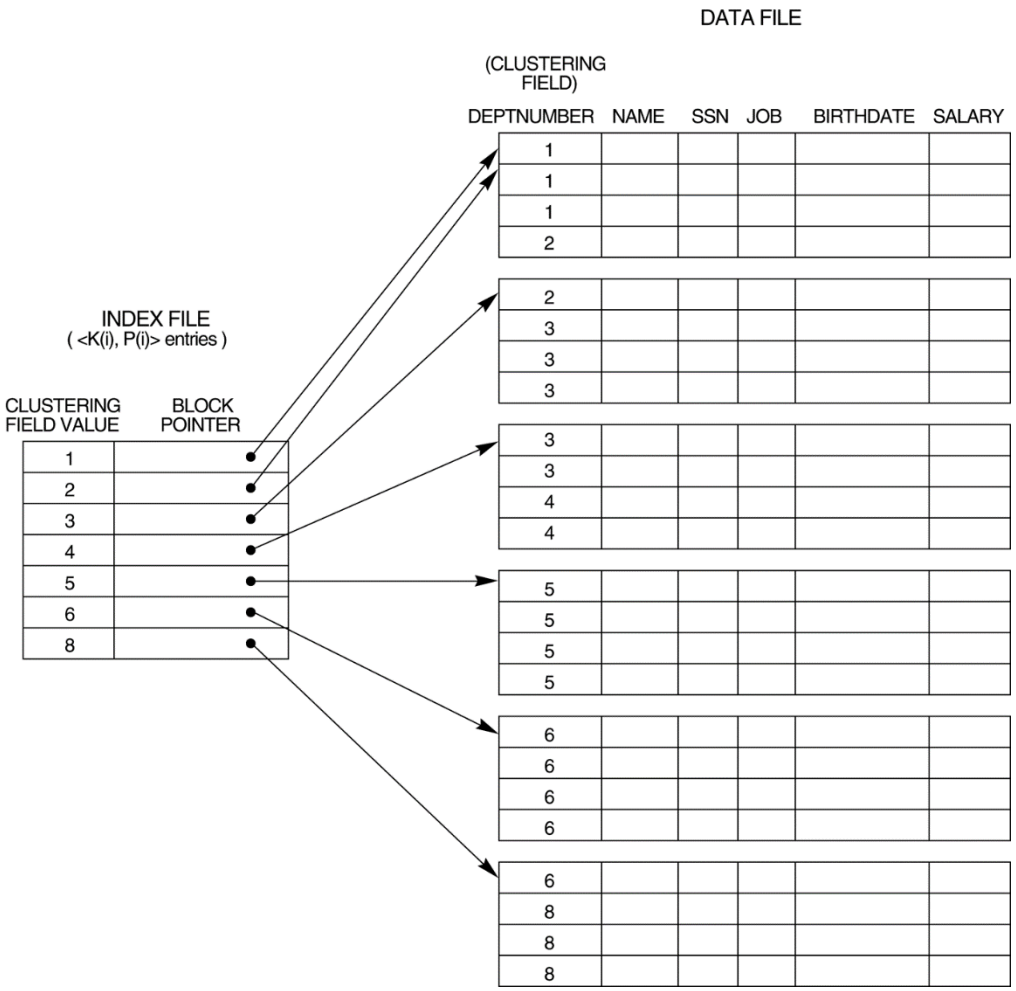

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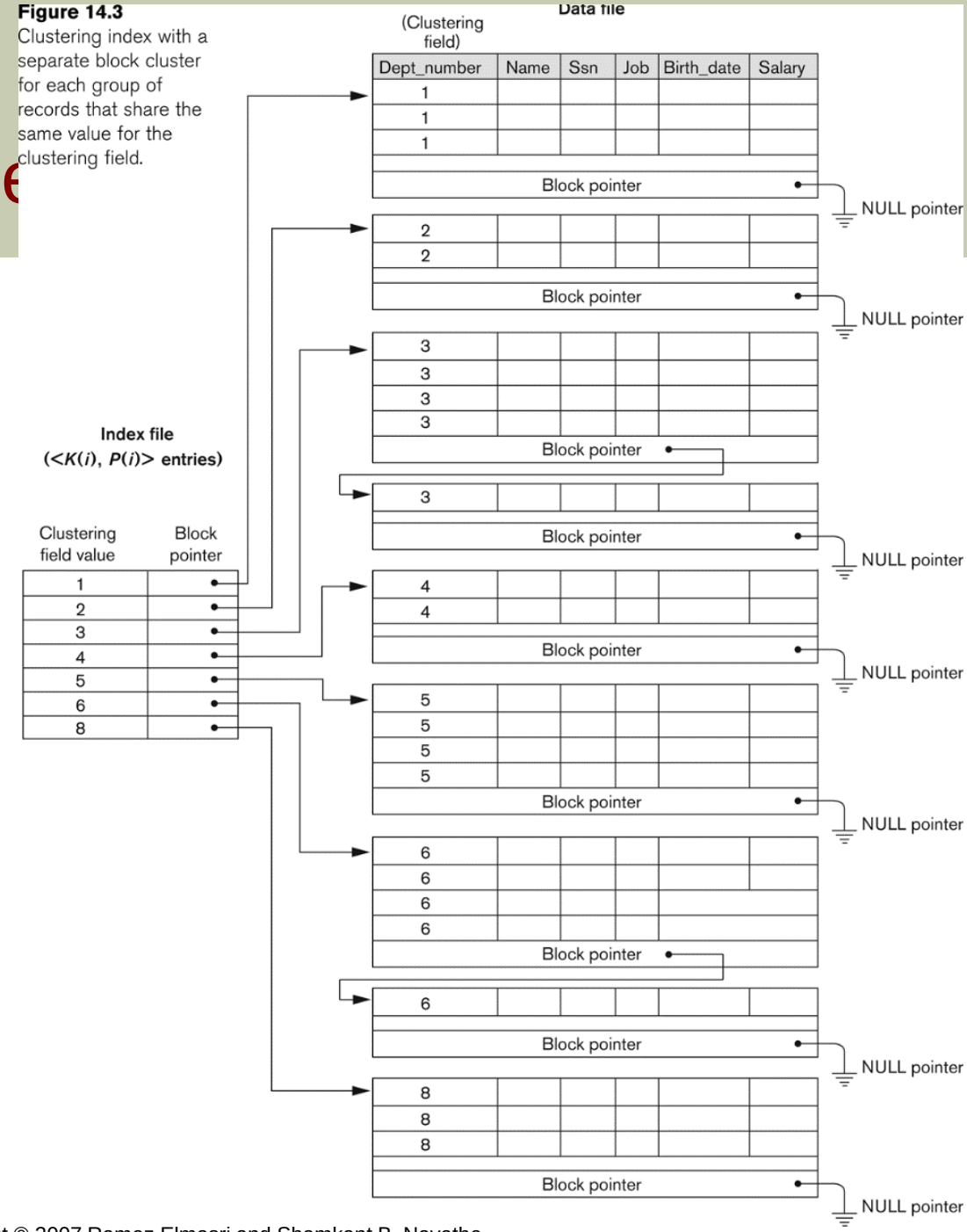
Figure 14.3
Clustering index with a separate block cluster for each group of records that share the same value for the clustering field.



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Types of Single-Level Indexes

- Secondary Index
  - A secondary index provides a secondary means of accessing a file for which some primary access already exists.
  - The secondary index may be on a field which is a candidate key and has a **unique value** in every record, or a **non-key with duplicate values**.
  - The index is an ordered file with two fields.
    - The first field is of the same data type as some **non-ordering field** of the data file that is an indexing field.
    - The second field is either a **block** pointer or a record pointer.
    - There can be *many* secondary indexes (and hence, indexing fields) for the same file.
  - Includes one entry *for each record* in the data file; hence, it is a *dense index*

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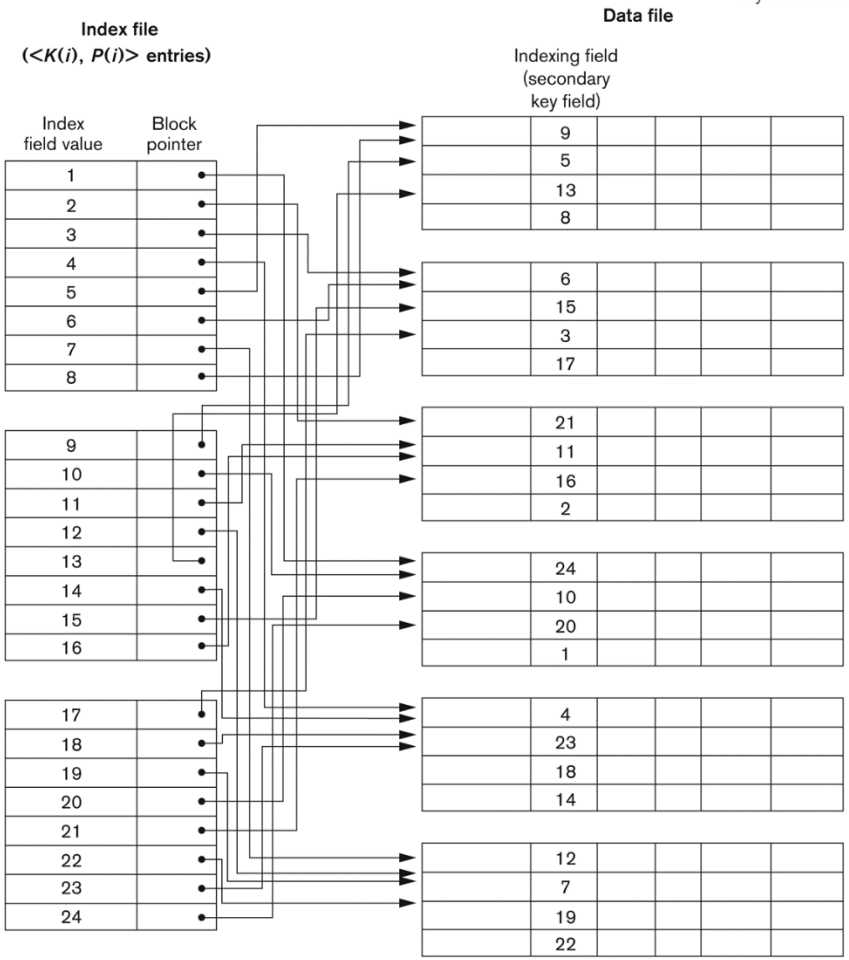
Example of a Dense Secondary Index

Figure 14.4
A dense secondary index (with block pointers) on a nonordering key field of a file.



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An Example of a Secondary Index

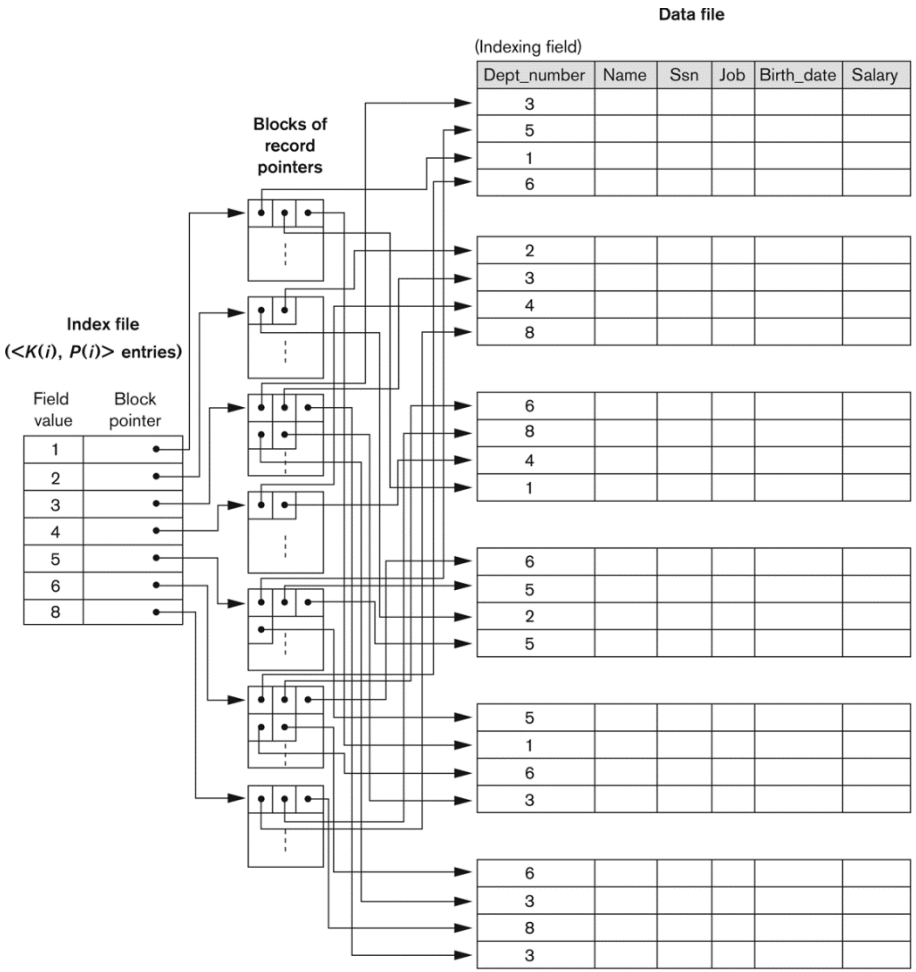

Figure 14.5
A secondary index (with record pointers) on a nonkey field implemented using one level of indirection so that index entries are of fixed length and have unique field values.

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Properties of Index Types

TABLE 14.2 PROPERTIES OF INDEX TYPES

TYPE OF INDEX	NUMBER OF (FIRST-LEVEL) INDEX ENTRIES	DENSE OR NONDENSE	BLOCK ANCHORING ON THE DATA FILE
Primary	Number of blocks in data file	Nondense	Yes
Clustering	Number of distinct index field values	Nondense	Yes/no[a]
Secondary (key)	Number of records in data file	Dense	No
Secondary (nonkey)	Number of records[b] or Number of distinct index field values[c]	Dense or Nondense	No

[a]Yes if every distinct value of the ordering field starts a new block; no otherwise.
[b]For option 1.
[c]For options 2 and 3.

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Multi-Level Indexes

- Because a single-level index is an ordered file, we can create a primary index *to the index itself*;
  - In this case, the original index file is called the ***first-level index*** and the index to the index is called the ***second-level index***.
- We can repeat the process, creating a third, fourth, ..., top level **until all entries of the *top level* fit in one disk block**
- A multi-level index can be created for any type of first-level index (primary, secondary, clustering) as long as the first-level index **consists of *more than one* disk block**

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A Two-level Primary Index

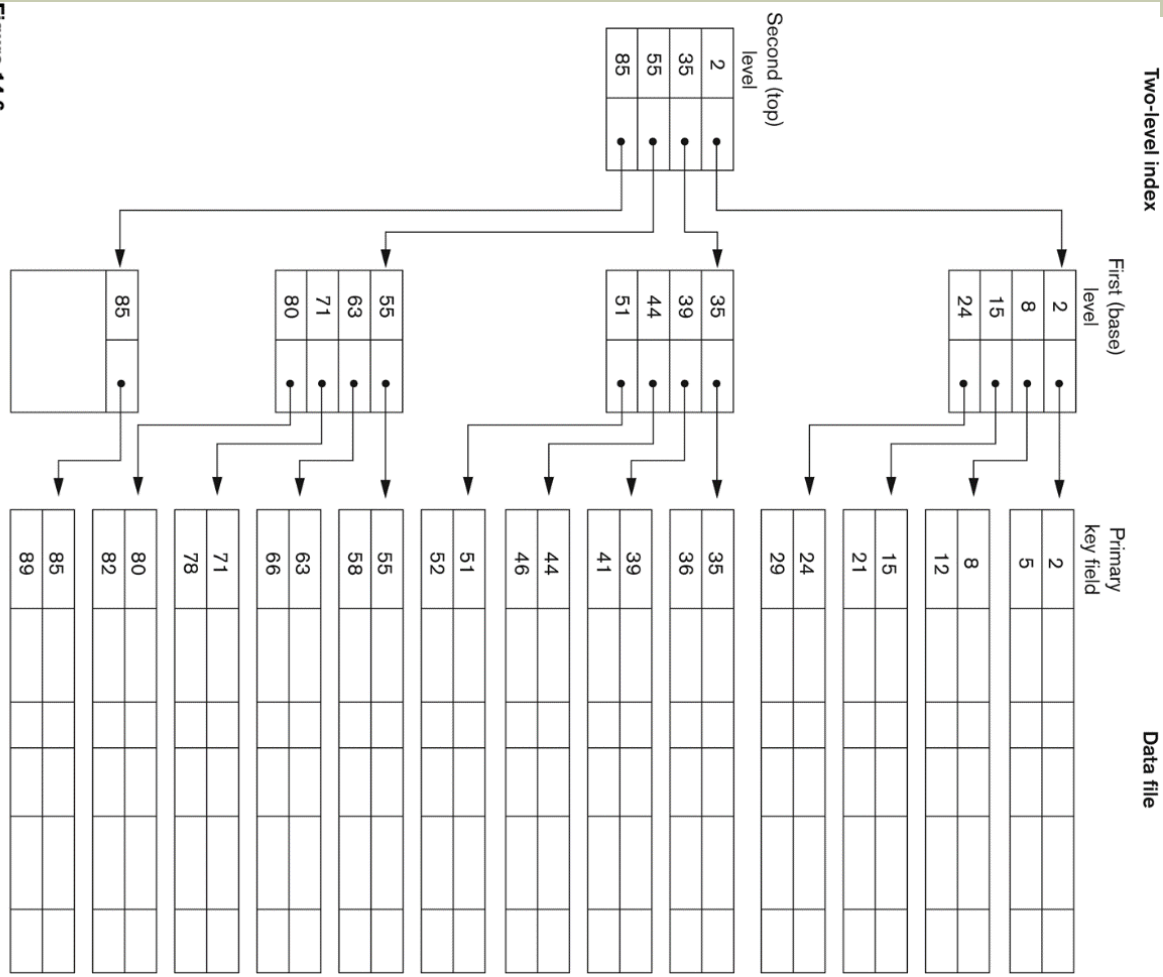

Figure 14.6
A two-level primary index resembling ISAM (Index Sequential Access Method) organization.

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Multi-Level Indexes

- Such a multi-level index is a form of ***search tree***
  - However, insertion and deletion of new index entries is a severe problem because every level of the index is an *ordered file*.

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A Node in a Search Tree with Pointers to Subtrees below It

- FIGURE 14.8

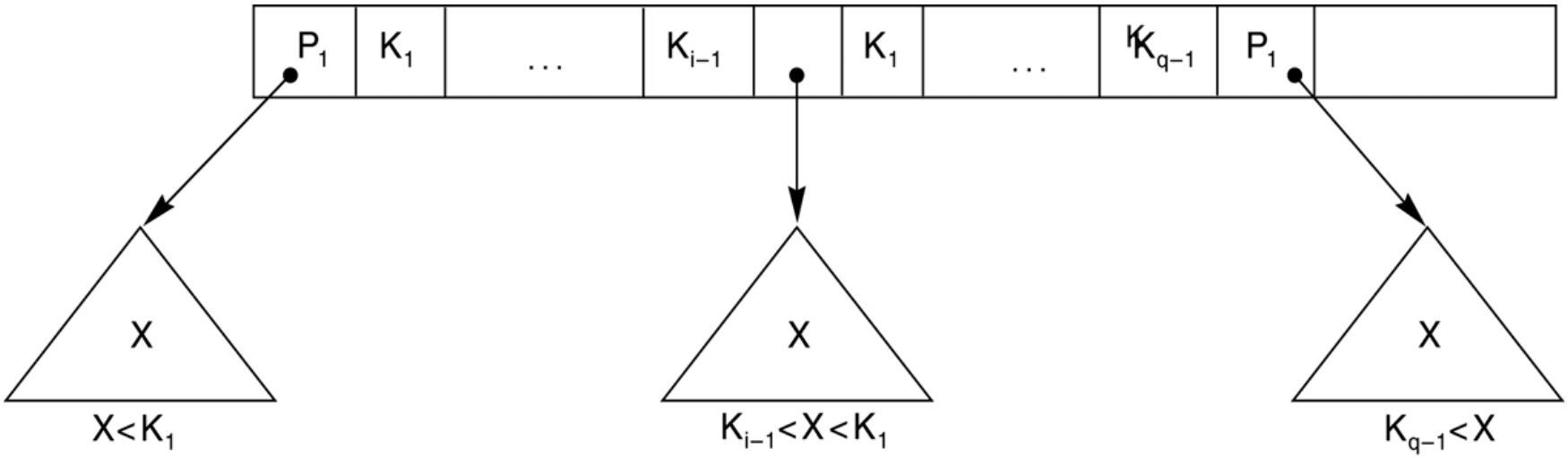

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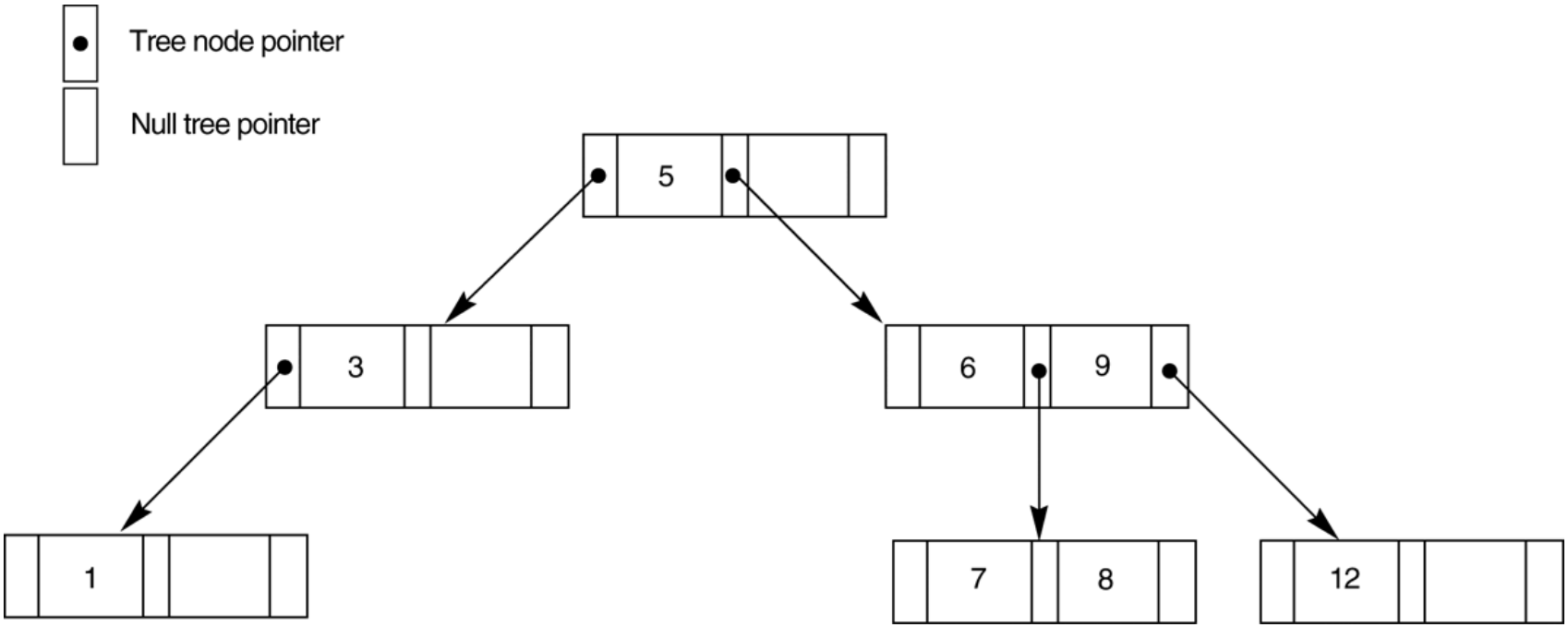
FIGURE 14.9
A search tree of order p = 3



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Dynamic Multilevel Indexes Using B-Trees and B+-Trees

- Most multi-level indexes use B-tree or B+-tree data structures because of the insertion and deletion problem
  - This leaves space in each tree node (disk block) to allow for new index entries
- These data structures are **variations of search trees** that allow efficient insertion and deletion of new search values.
- In B-Tree and B+-Tree data structures, **each node corresponds to a disk block**
- Each node is kept between half-full and completely full

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Dynamic Multilevel Indexes Using B-Trees and B+-Trees (contd.)

- An insertion into a node that is not full is quite efficient
  - If a node is full the insertion causes a split into two nodes
- Splitting may propagate to other tree levels
- A deletion is quite efficient if a node does not become less than half full
- If a deletion causes a node to become less than half full, it must be merged with neighboring nodes

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Difference between B-tree and B+-tree

- In a B-tree, pointers to data records exist at all levels of the tree
- In a B+-tree, all pointers to data records exists at the leaf-level nodes
- A B+-tree can have less levels (or higher capacity of search values) than the corresponding B-tree

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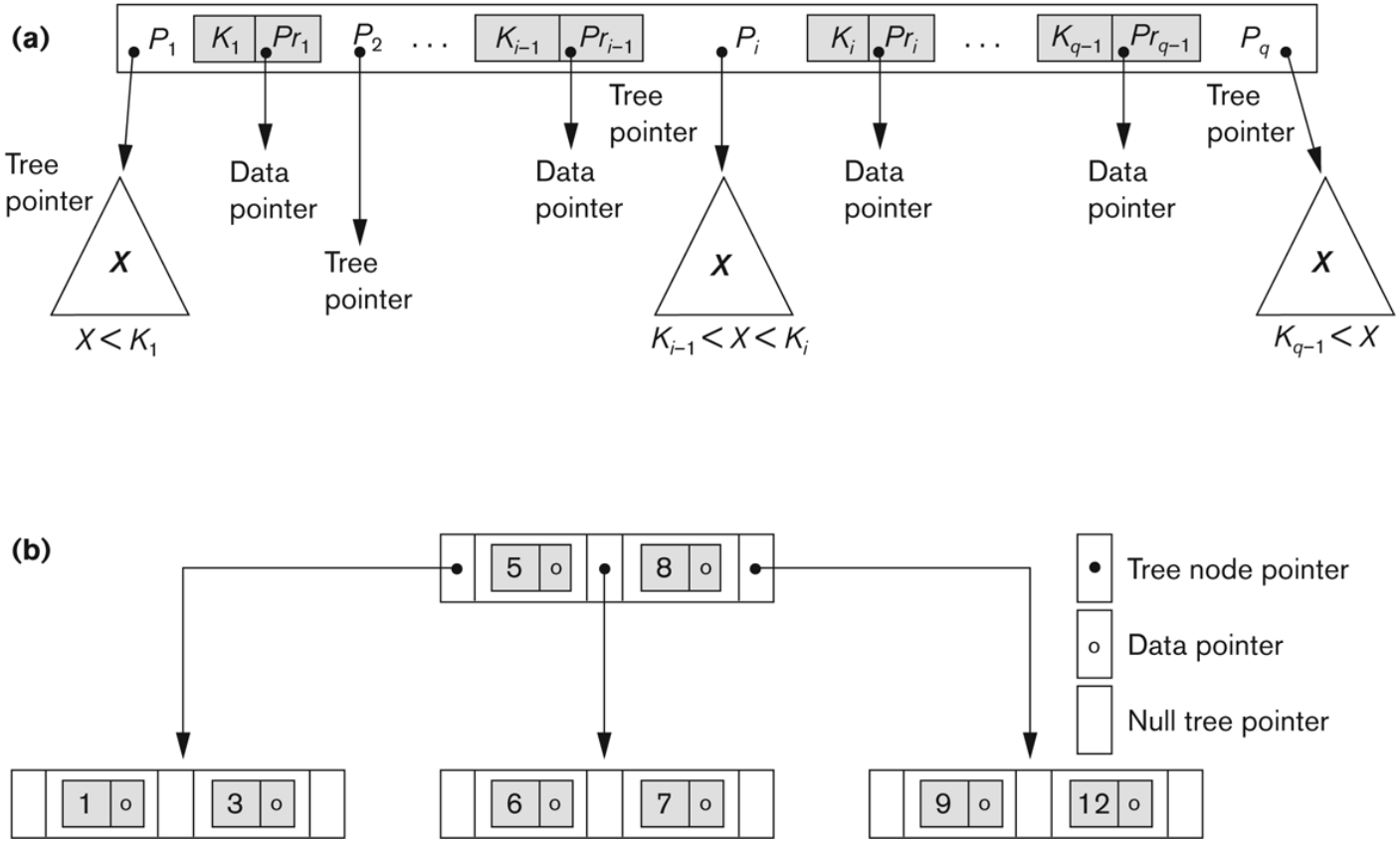
B-tree Structures

Figure 14.10
B-Tree structures. (a) A node in a B-tree with $q - 1$ search values. (b) A B-tree of order $p = 3$.The values were inserted in the order 8, 5, 1, 7, 3, 12, 9, 6.



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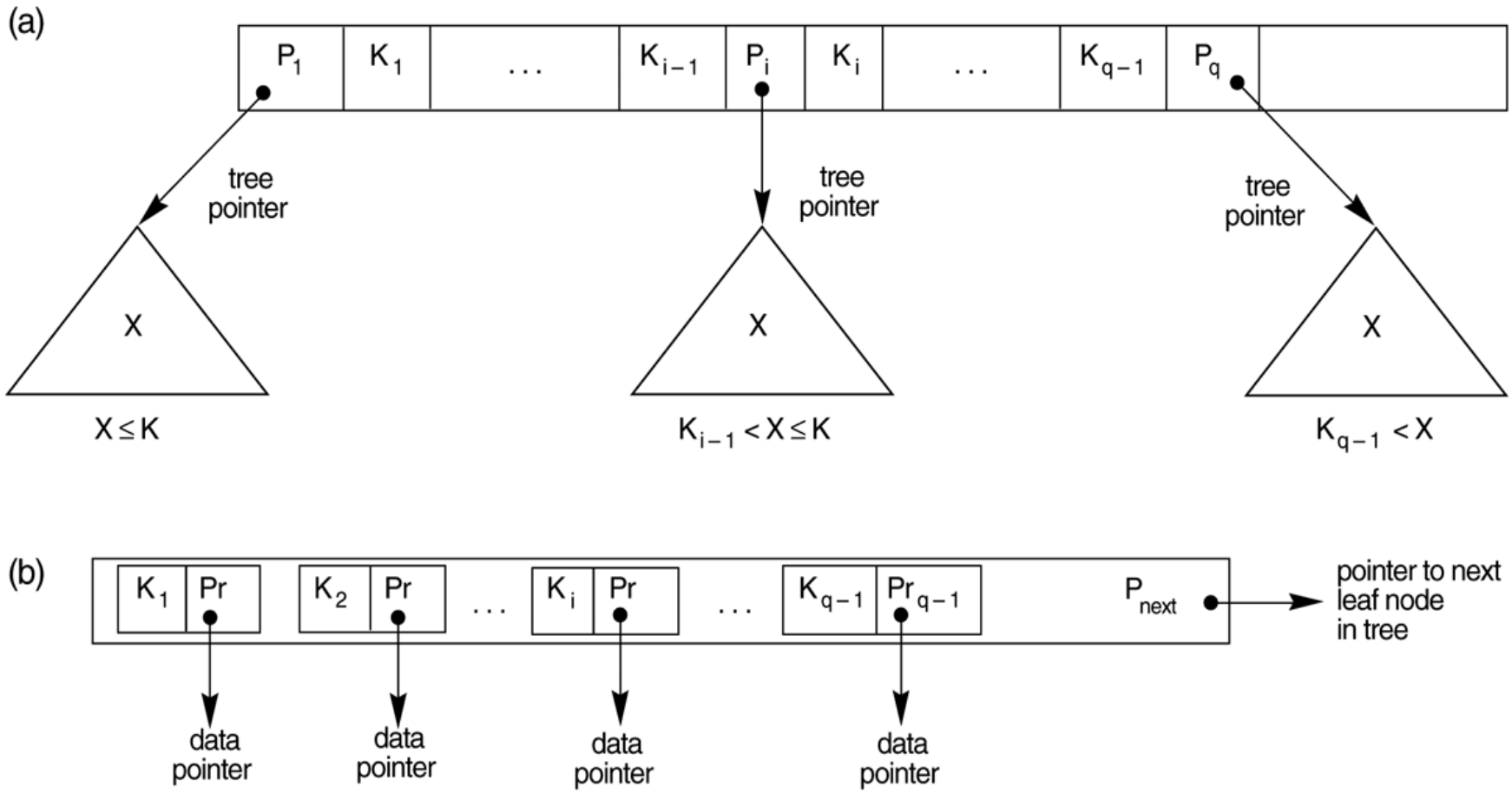
The Nodes of a B+-tree

- FIGURE 14.11 The nodes of a B+-tree
  - (a) Internal node of a B+-tree with q –1 search values.
  - (b) Leaf node of a B+-tree with q – 1 search values and q – 1 data pointers.



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An Example of an Insertion in a B+-tree

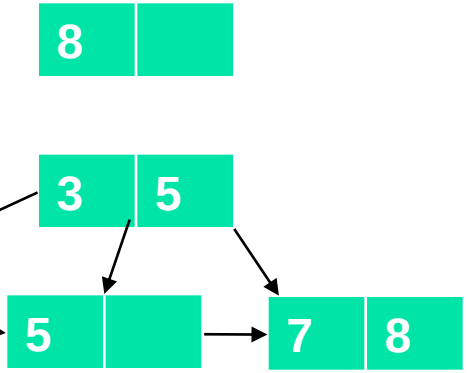
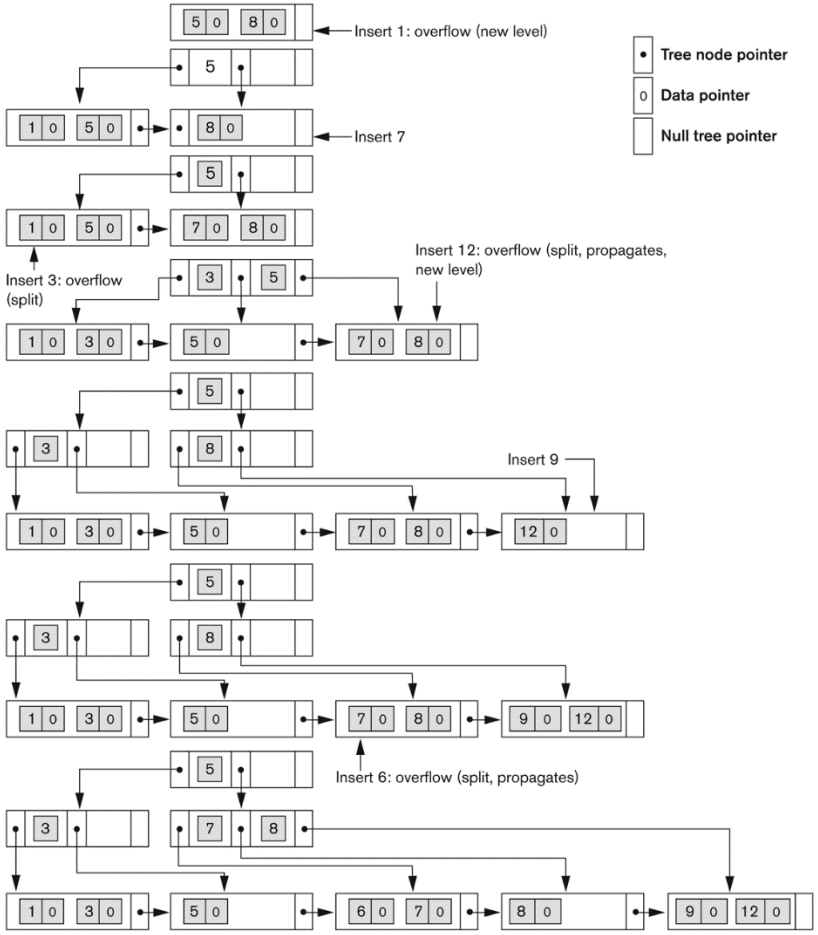


Figure 14.12
An example of insertion in a B+-tree with $p = 3$ and $p_{leaf} = 2$.

Insertion sequence: 8, 5, 1, 7, 3, 12, 9, 6



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An Example of a Deletion in a B+-tree

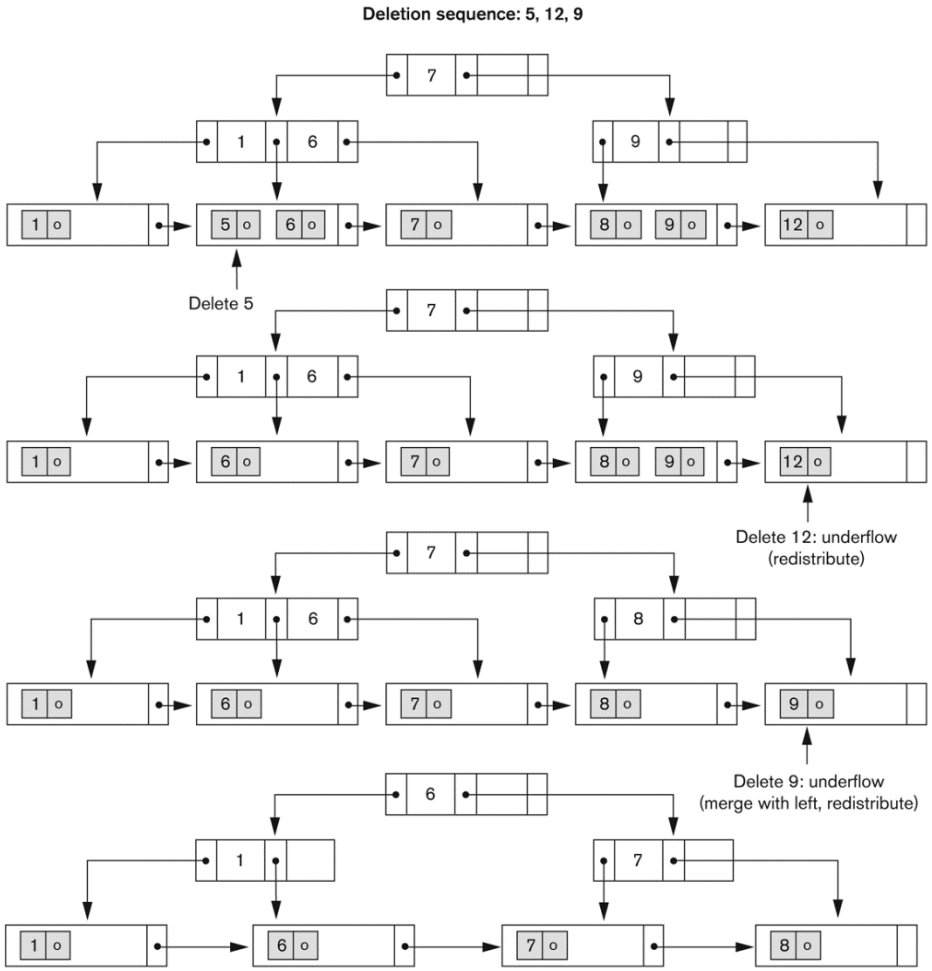

Figure 14.13
An example of deletion from a B+-tree.

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Summary

- Types of Single-level Ordered Indexes
  - Primary Indexes
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